

WAMIQ AJMAL ENGINEERING_PORTFOLIO

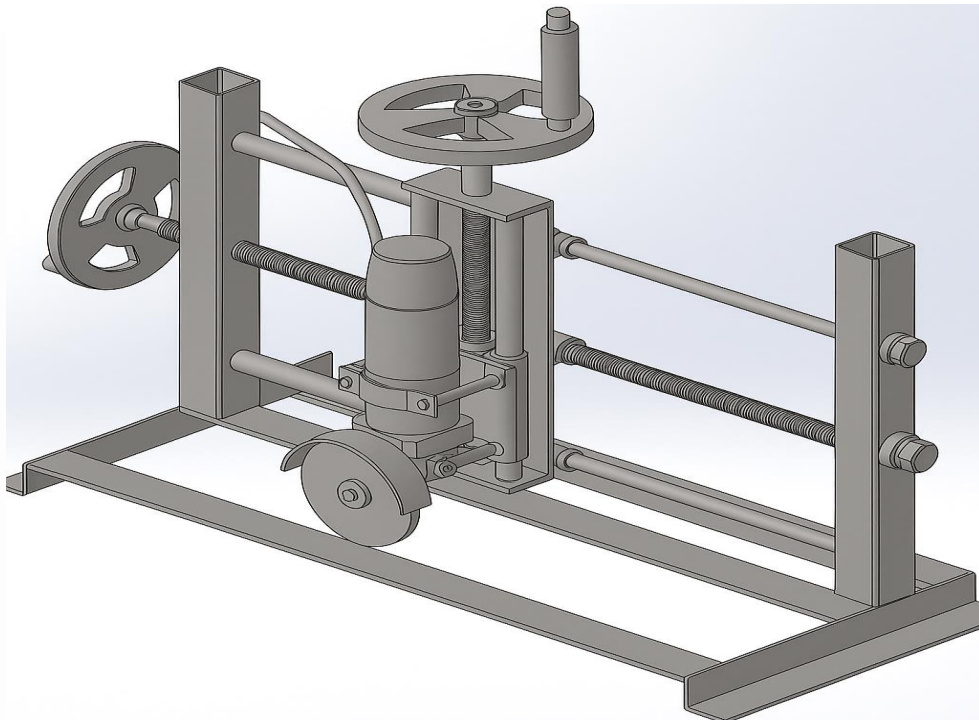
Comprehensive CAD Engineering & Technical Documentation

CORE COMPETENCIES

- ✓ Complex Assembly Modeling
- ✓ Industrial Mechanism Design
- ✓ Sheet Metal & Weldments
- ✓ Parametric Part Creation
- ✓ Technical Drafting & GD&T
- ✓ Tolerance Analysis

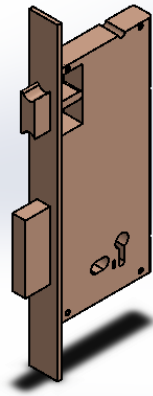
PRIMARY MACHINE ASSEMBLIES

Portable Metal Cutting Machine (PMCM)



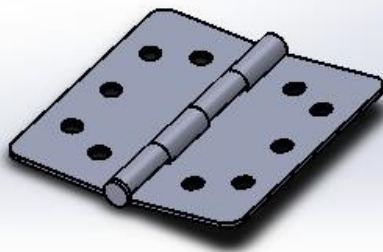
Custom-engineered multi-axis cutting assembly featuring precision lead screw adjustments and dual-rod support for high stability during operation.

Main Door Plate

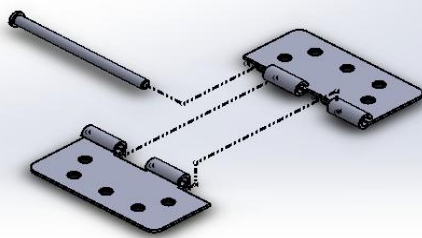


Custom door surface plate featuring precision apertures and mounting configurations.

Industrial Hinged System

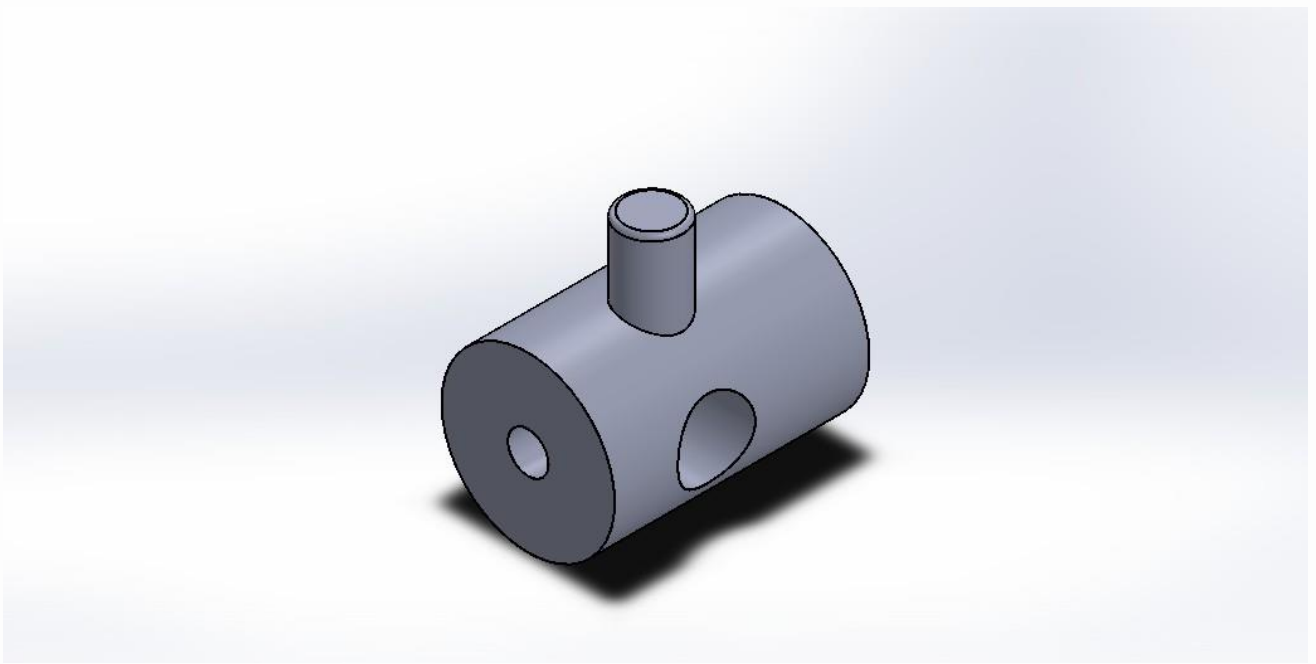


Heavy-duty dual-plate hinge assembly designed for high-cycle industrial access panels.



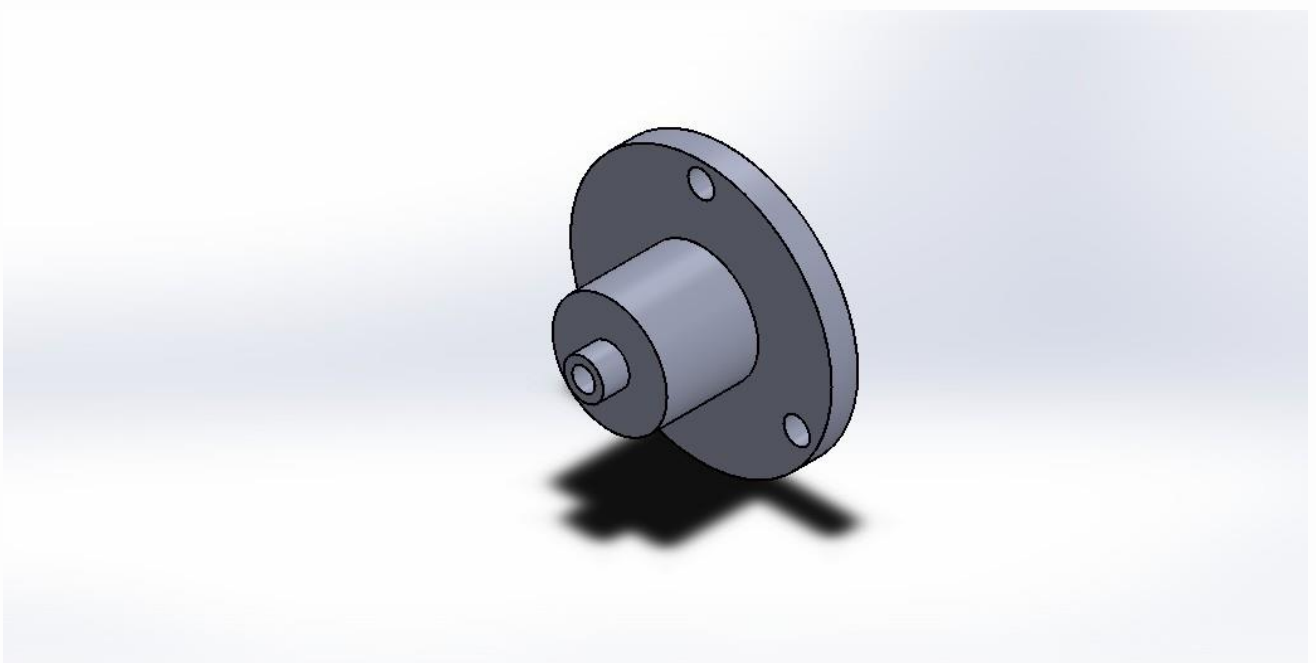
DOOR LOCK & SECURITY SUB-SYSTEMS

Main Door Lock Assembly



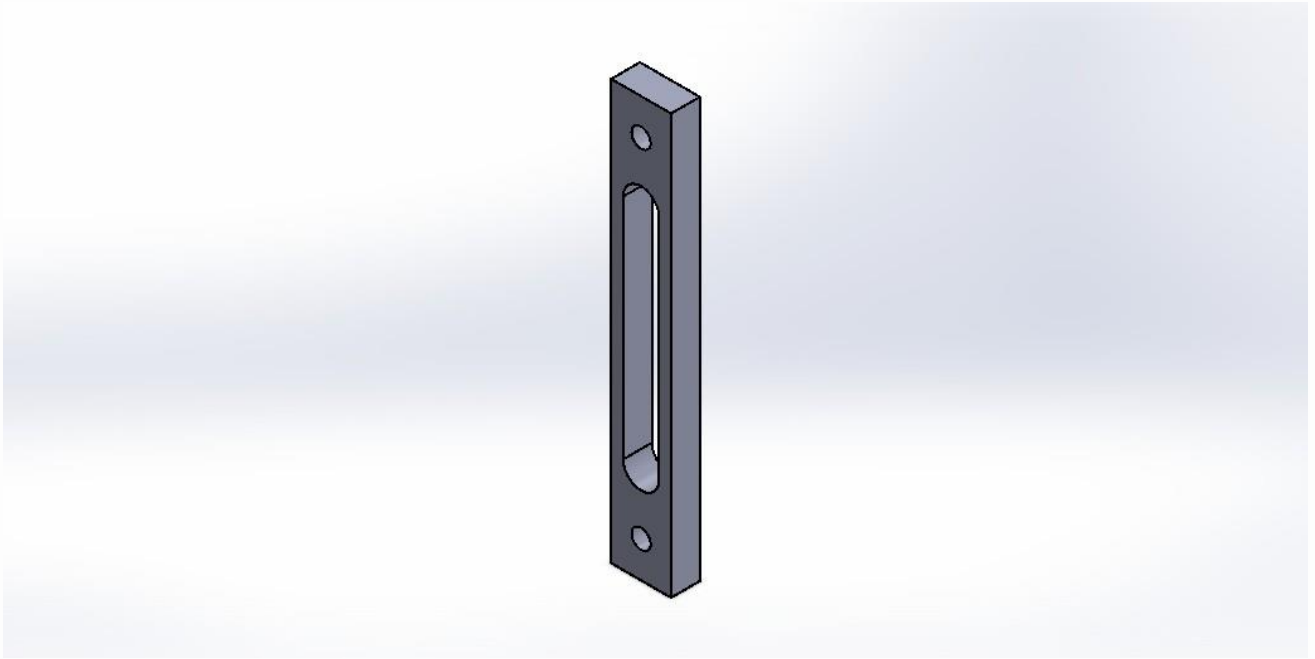
Integrated locking mechanism including linear rod actuators for multi-point security.

Lock Handle Holder



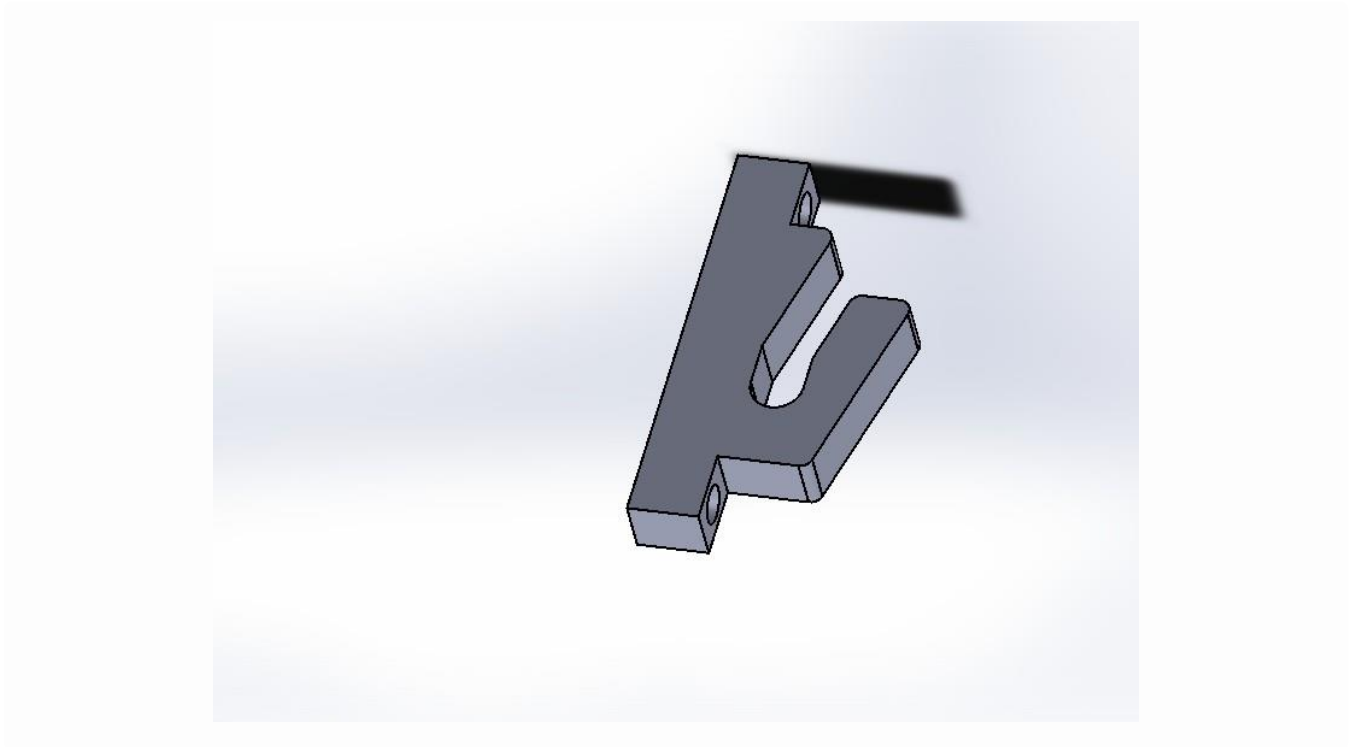
Precision mounting component for the ergonomic handle interface.

Hard Point Attachment



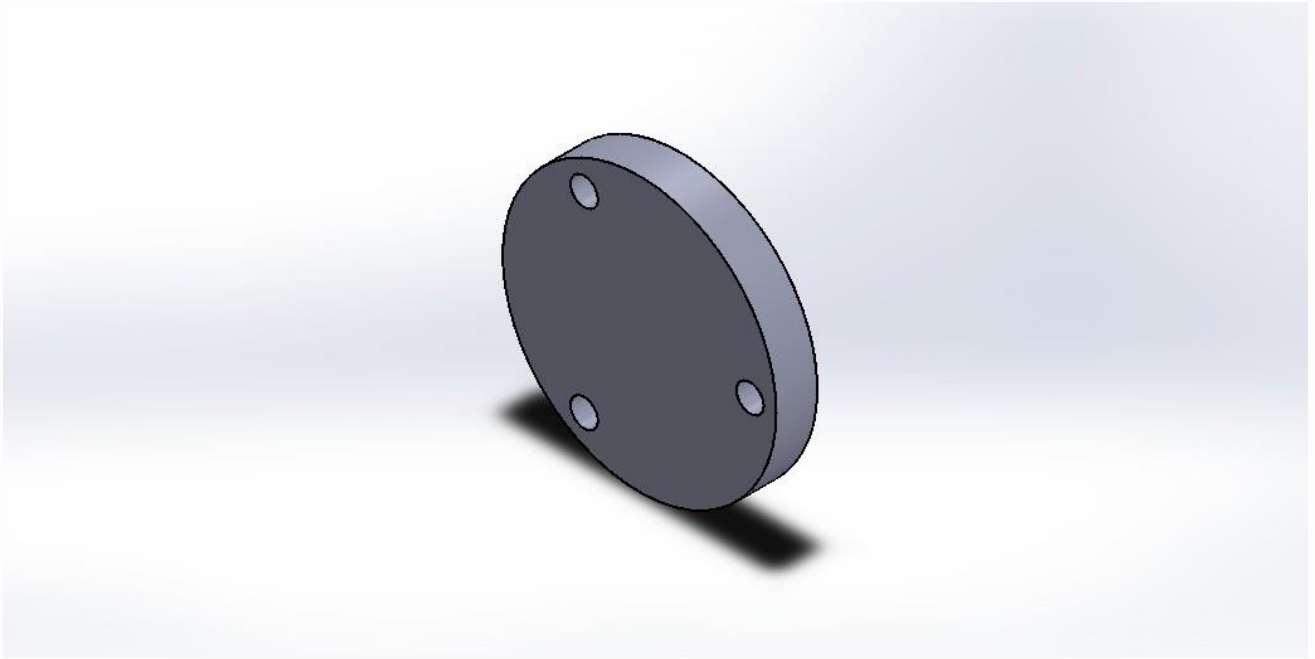
Reinforcement hub designed for secure mechanical anchoring of the lock assembly.

Roller Pin Holder



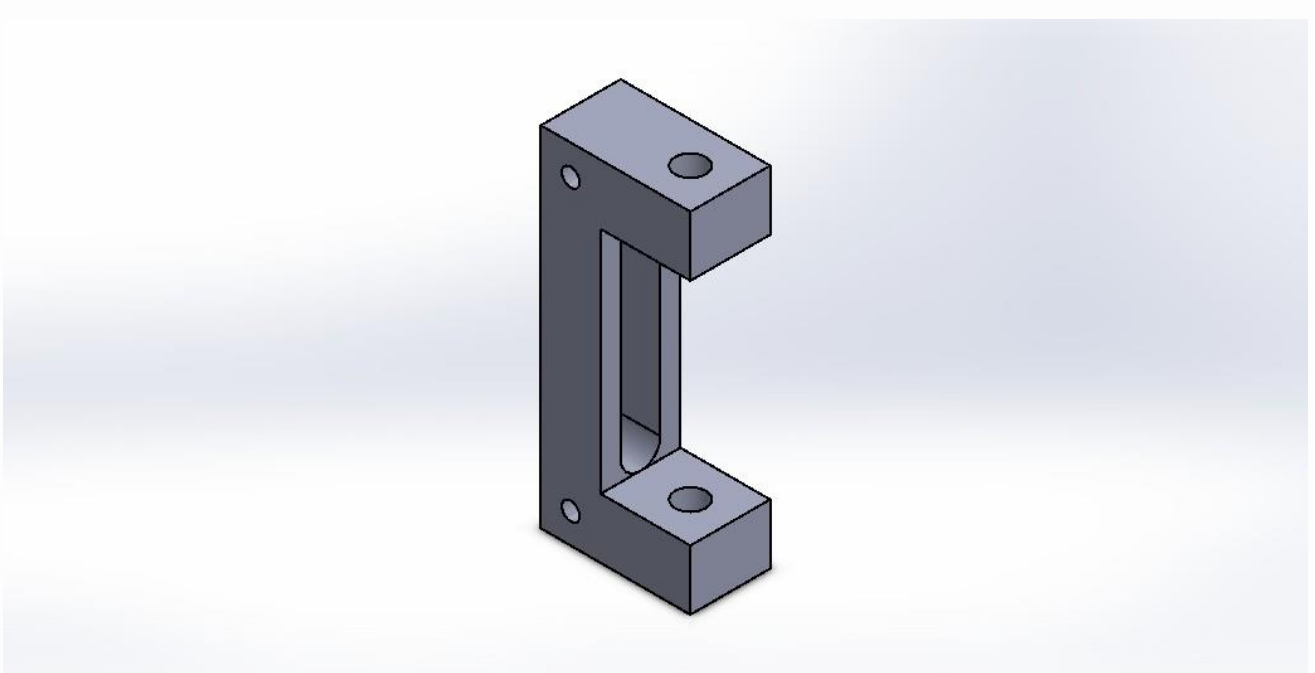
Specialized guide assembly for smooth, friction-reduced door latching.

Handle Washer Component



Surface protection and rotation spacer for the door handle mechanism.

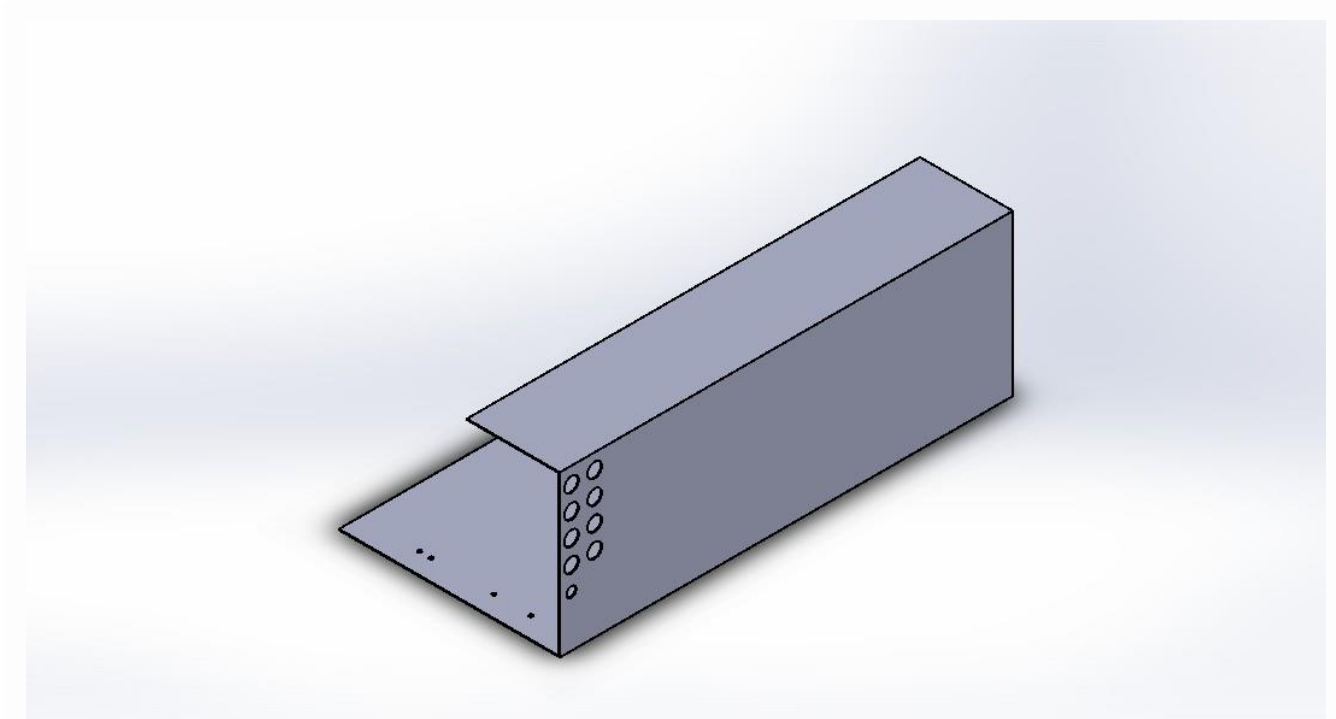
Lock Rod Guide



Alignment bracket ensuring precise linear motion of locking rods.

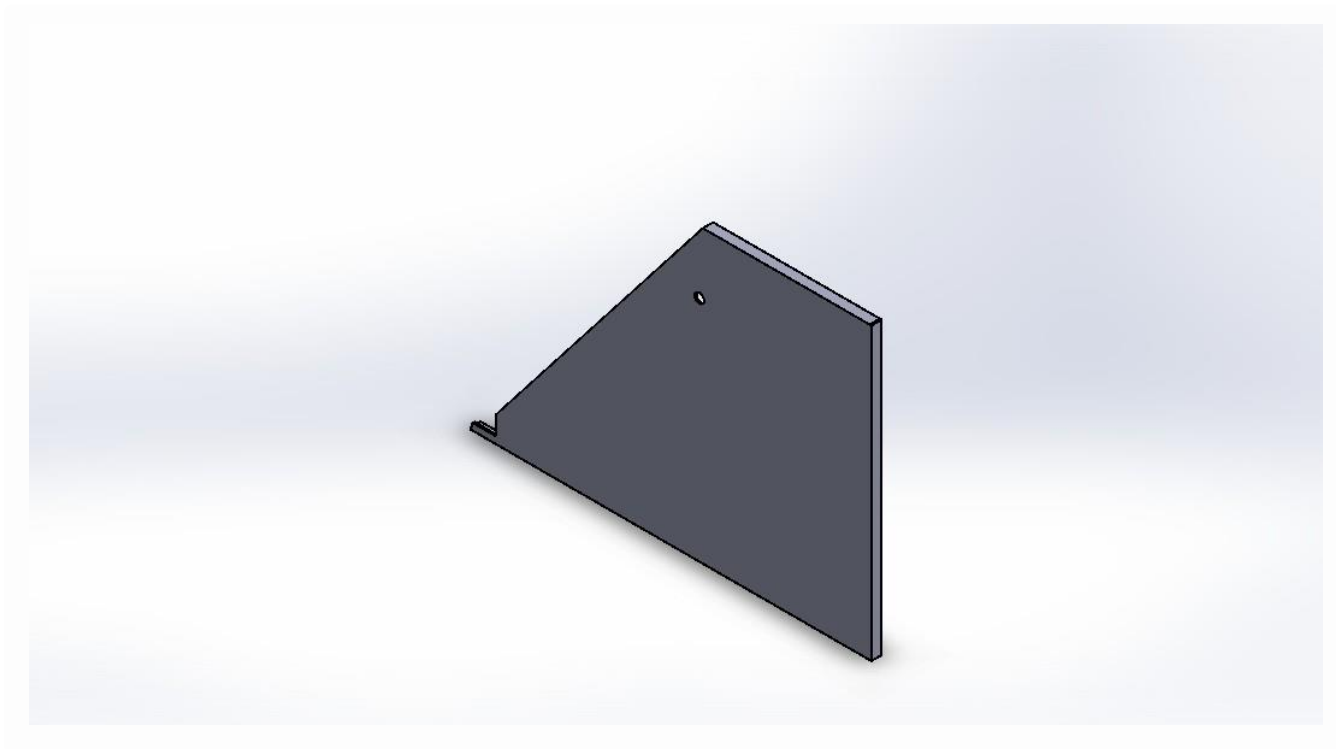
MAIN BODY & STRUCTURAL CHASSIS

Main Body Sheet (Aperture Pattern)



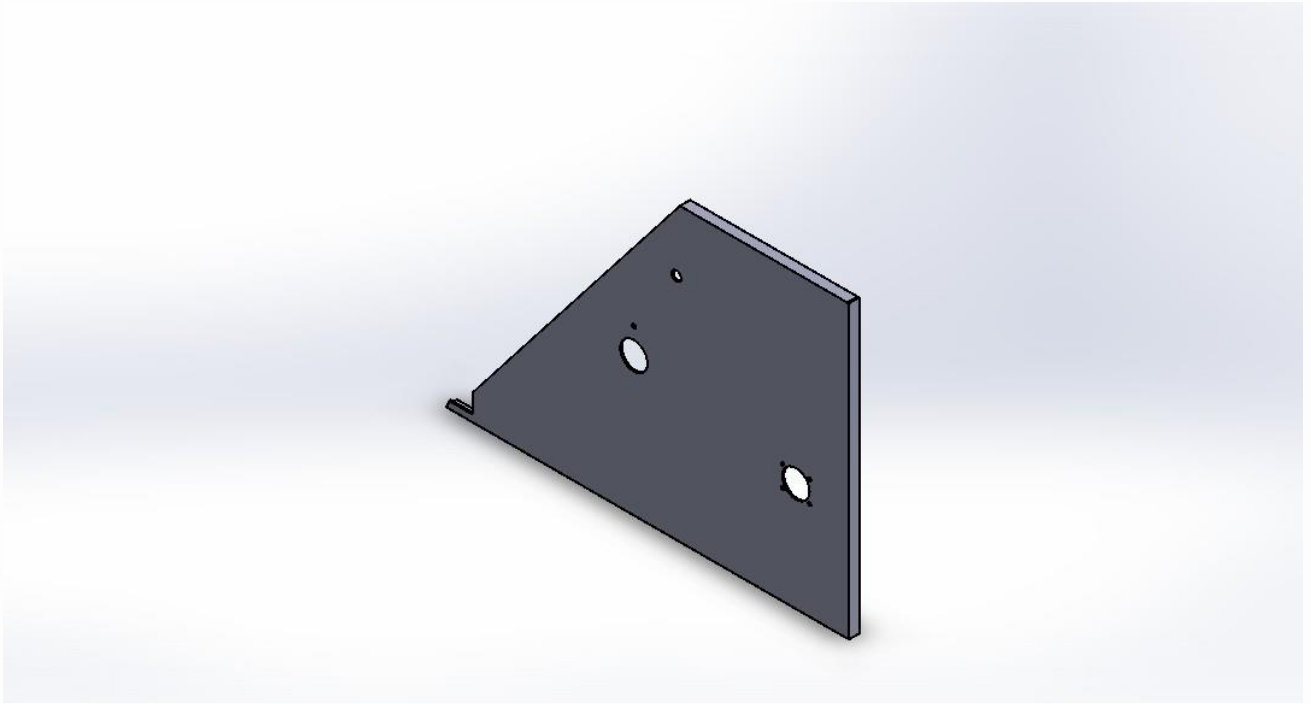
Primary chassis component with integrated ventilation and mounting apertures.

Chassis Side Sheet - Variant 1



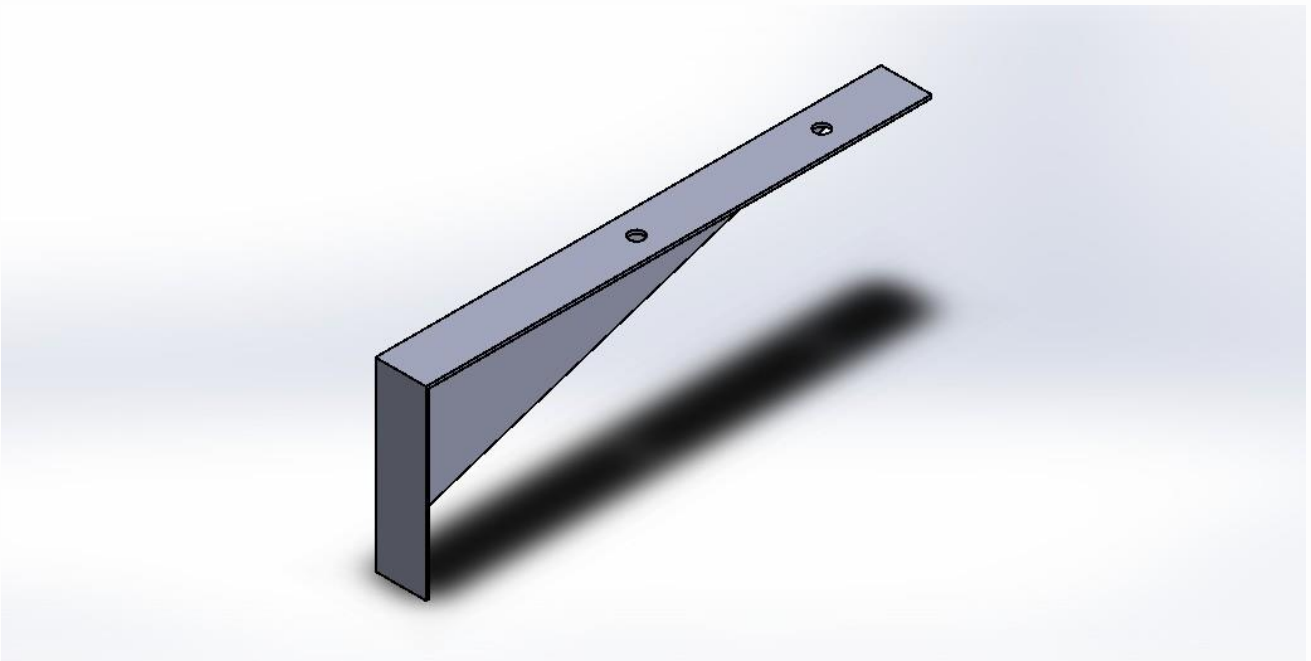
Structural reinforcement panel for the main body housing.

Chassis Side Sheet - Variant 2



Complementary side panel featuring unique slot patterns for modular part integration.

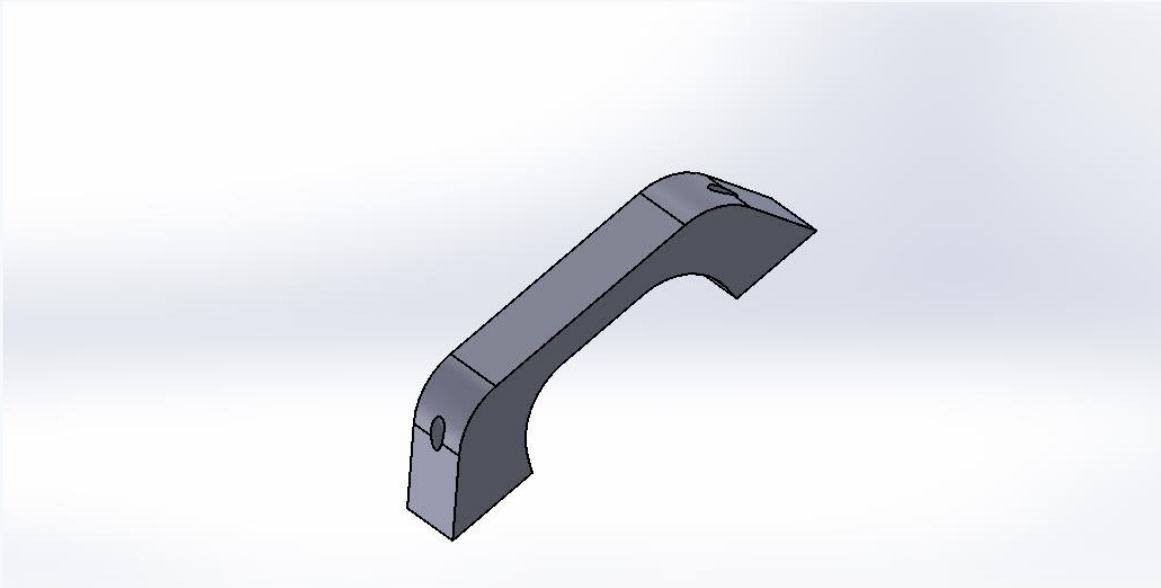
Mounting Bracket Assembly



Bilateral support system for internal equipment housing (CJ Box).

Automotive & Interior Component Design

Car Interior Handle



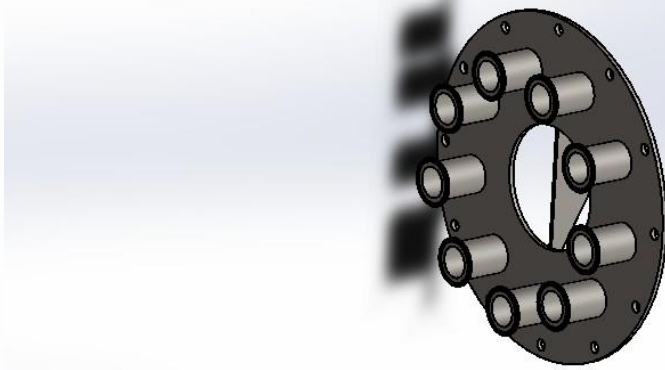
Ergonomic automotive interior handle design featuring complex surface contours and integrated mounting points.

Battery Clamp Assembly



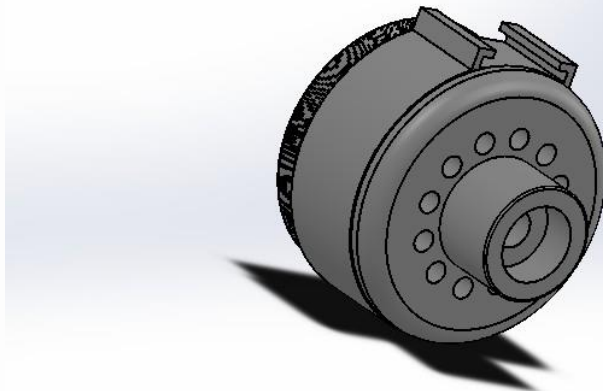
Securing frame designed for vibration-resistant internal power storage.

8-Way Distribution Manifold



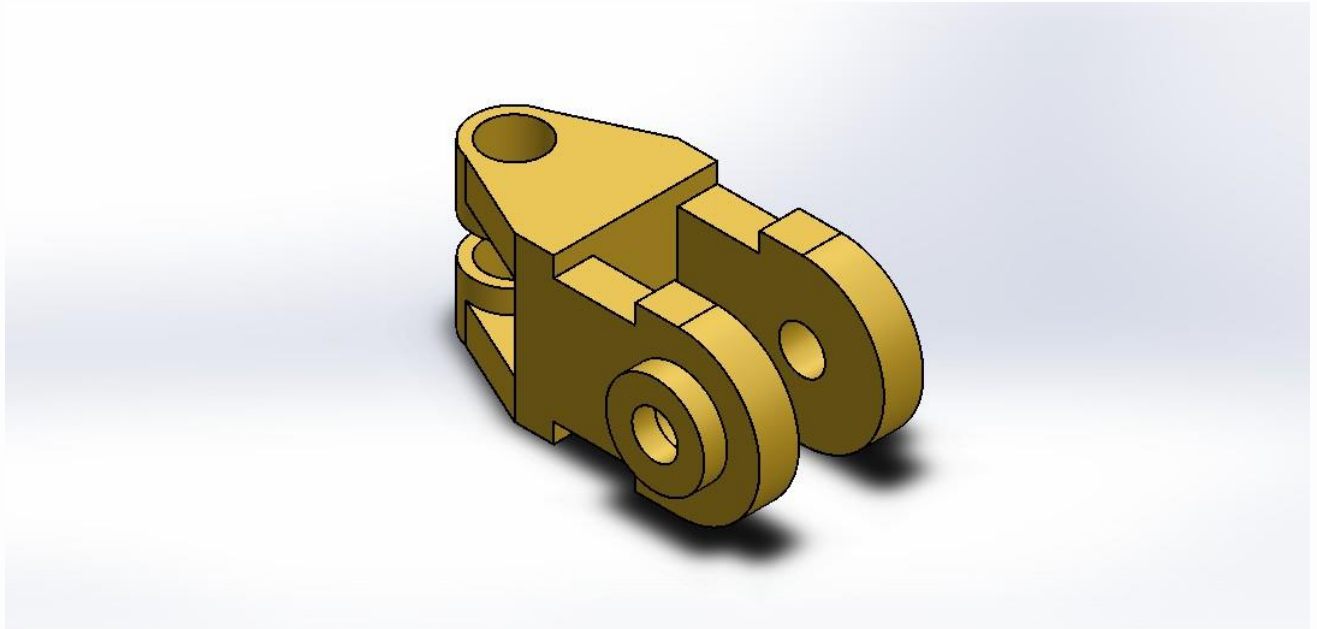
Complex distribution plate for multi-line pneumatic or fluid routing.

Step Fitting Cylinder



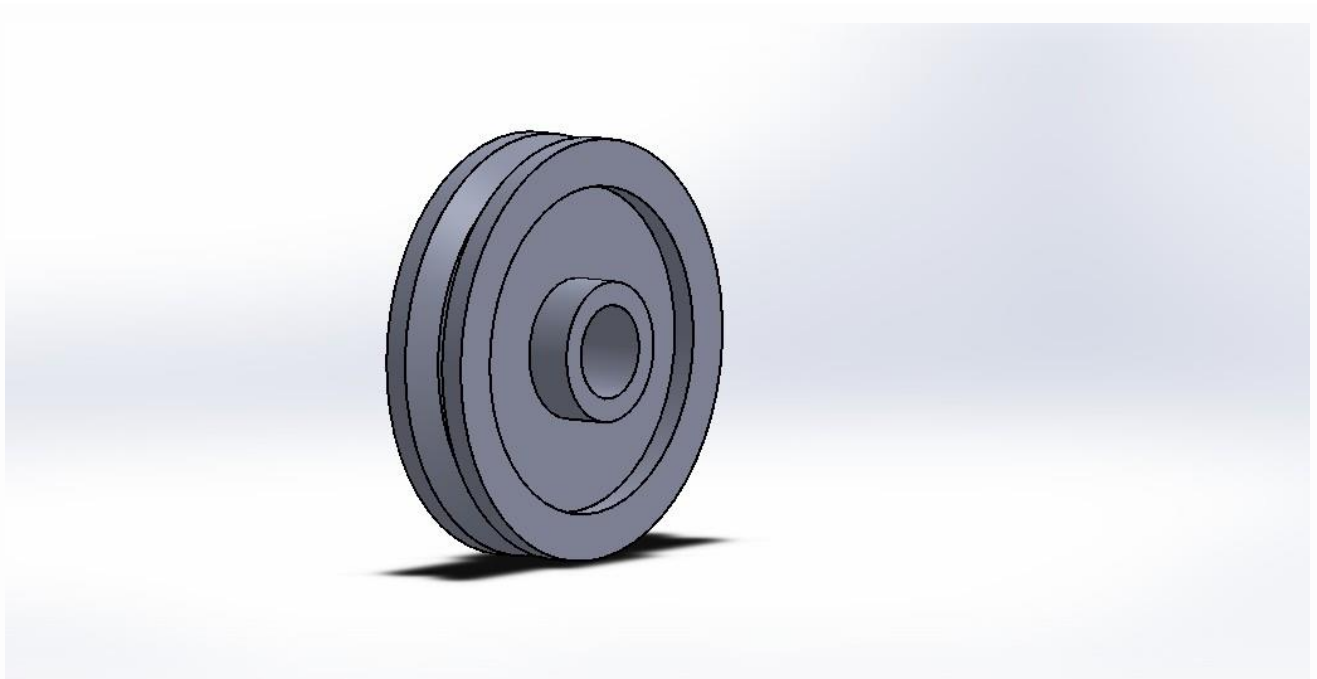
Tiered cylindrical component for specific mechanical fitting transitions.

High-Load Clevis Linkage



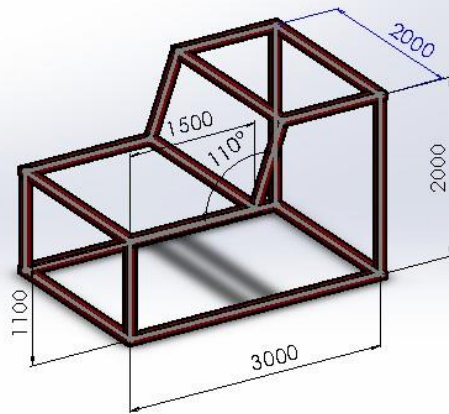
Robust pivot component designed for heavy-duty articulating joints.

V-Belt Transmission Pulley



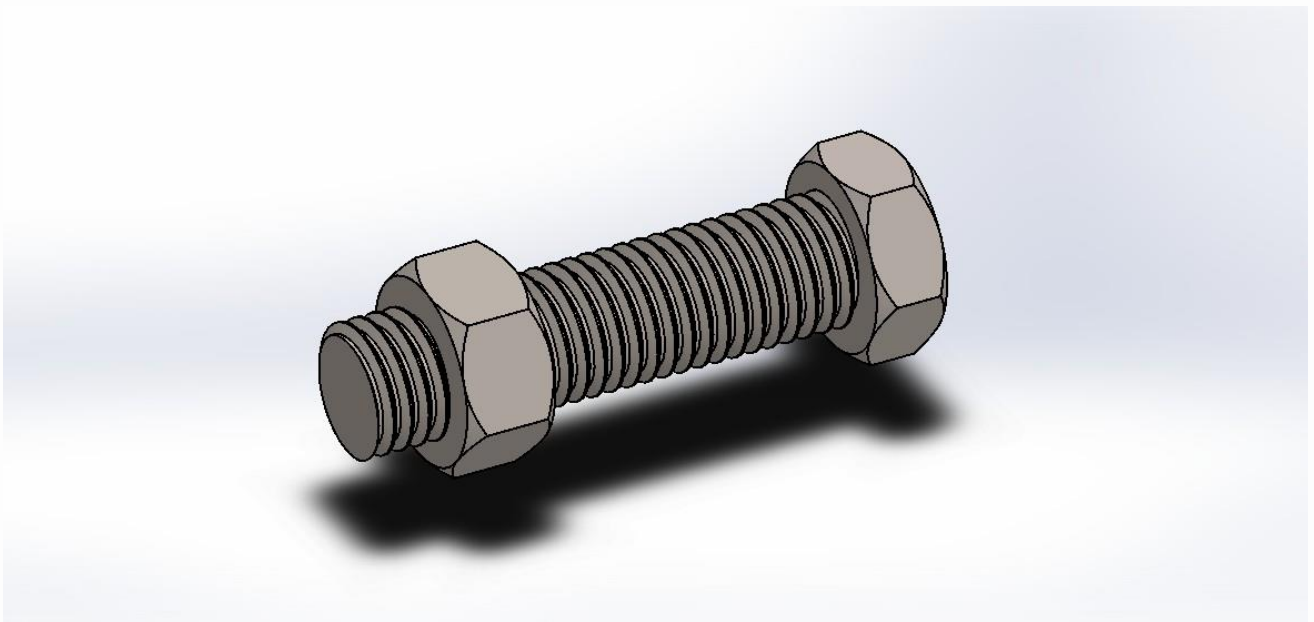
Precision-turned pulley with integrated central hub and keyway.

Structural 3D Layout



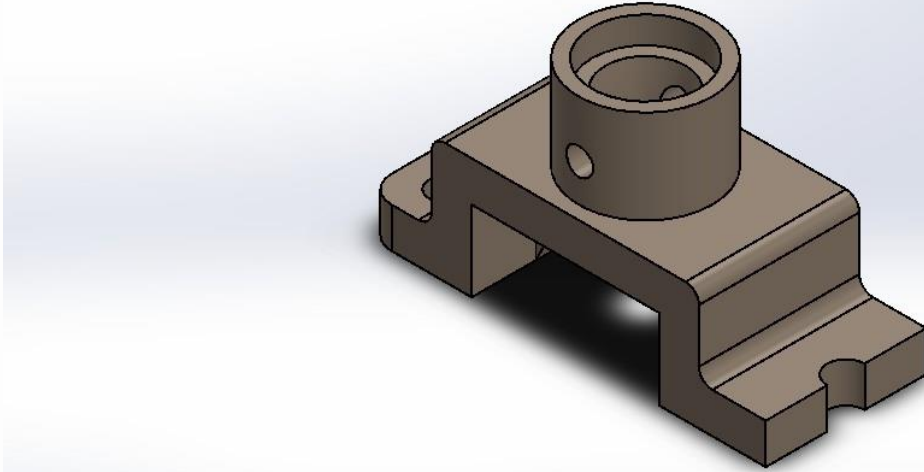
Industrial framework layout with specific angular (110°) and dimensional constraints.

Industrial Fastener Set



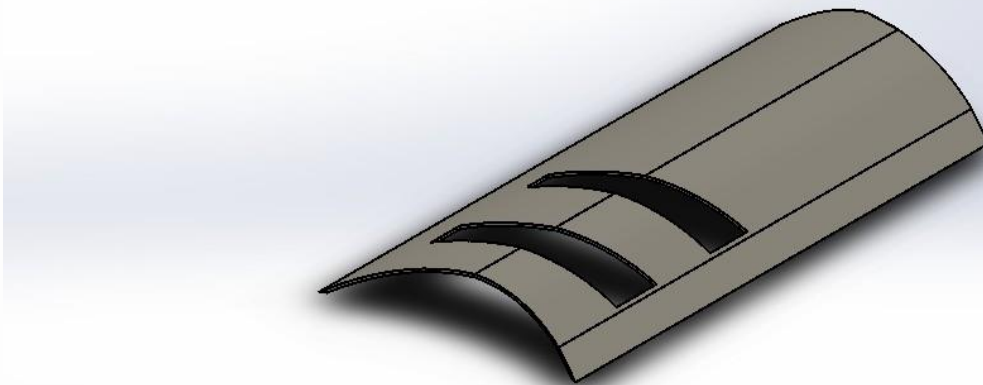
Standard nut and bolt assembly used for hardware clearance verification.

Solid Modeling Housing



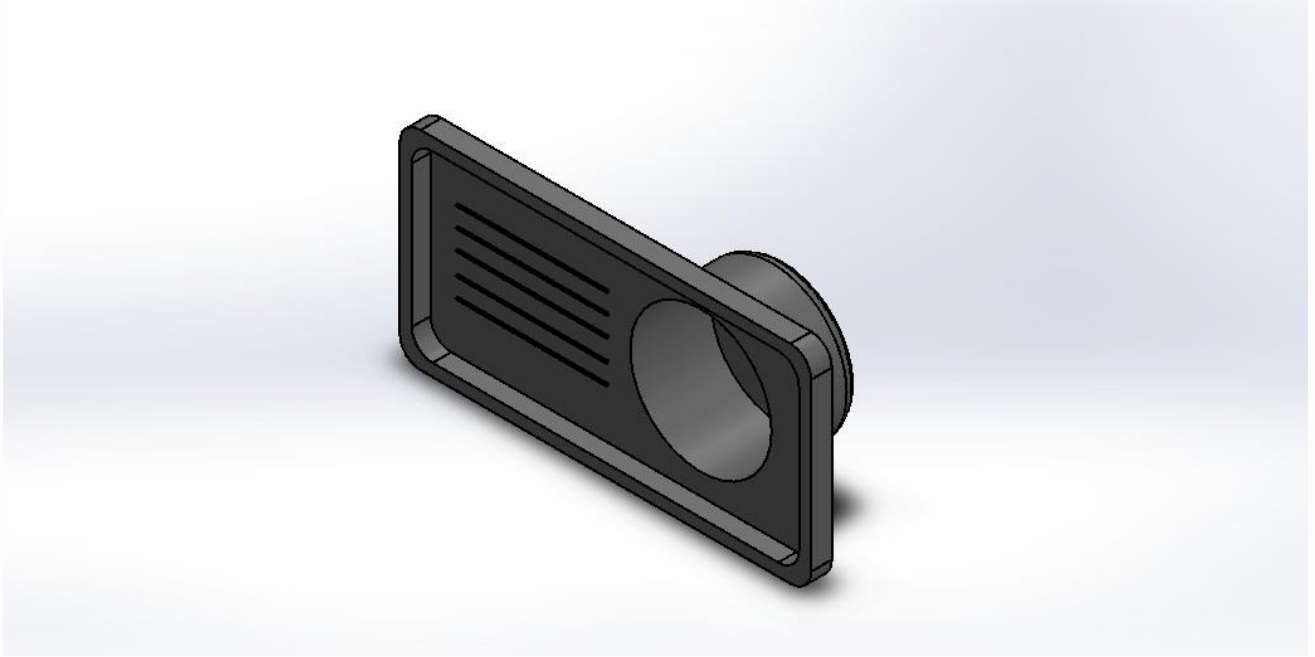
Advanced solid-body creation for integrated mechanical protective covers.

Revo Aerodynamic Mold



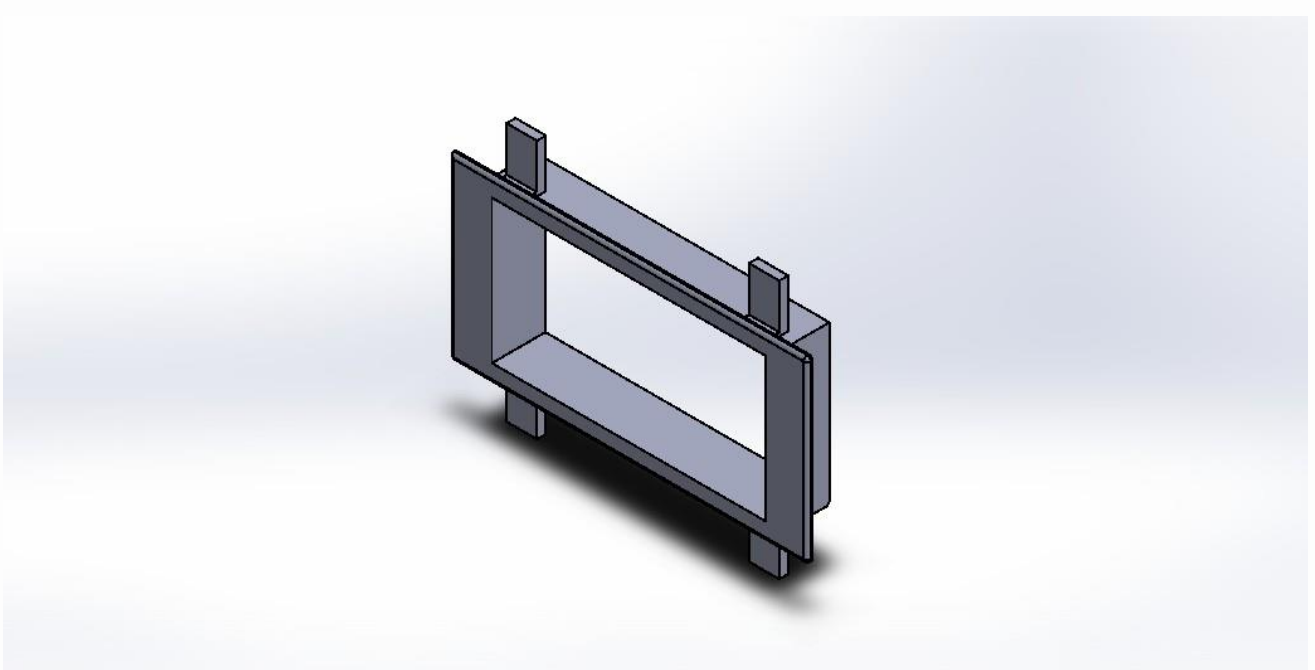
Surfacing project for light housings with complex double-curvature geometry.

Slotted Ventilation Bezel



Precision bezel design with heat dissipation slots.

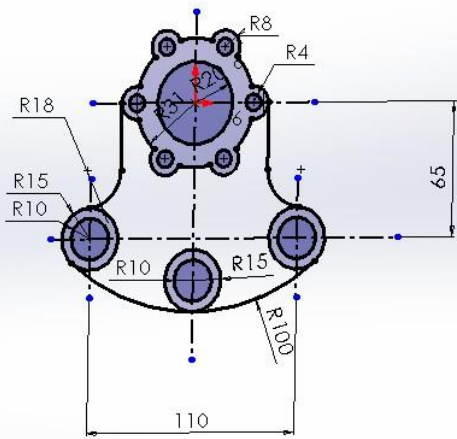
Modular Structural Frame



Lightweight spacer frame for maintaining internal alignment.

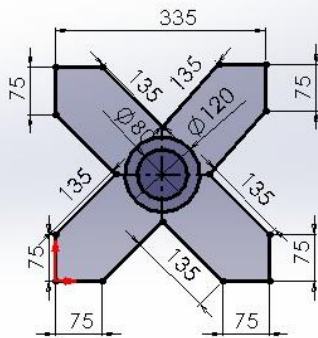
TECHNICAL DESIGN SKETCHES

Parametric 2D Constraint Logic



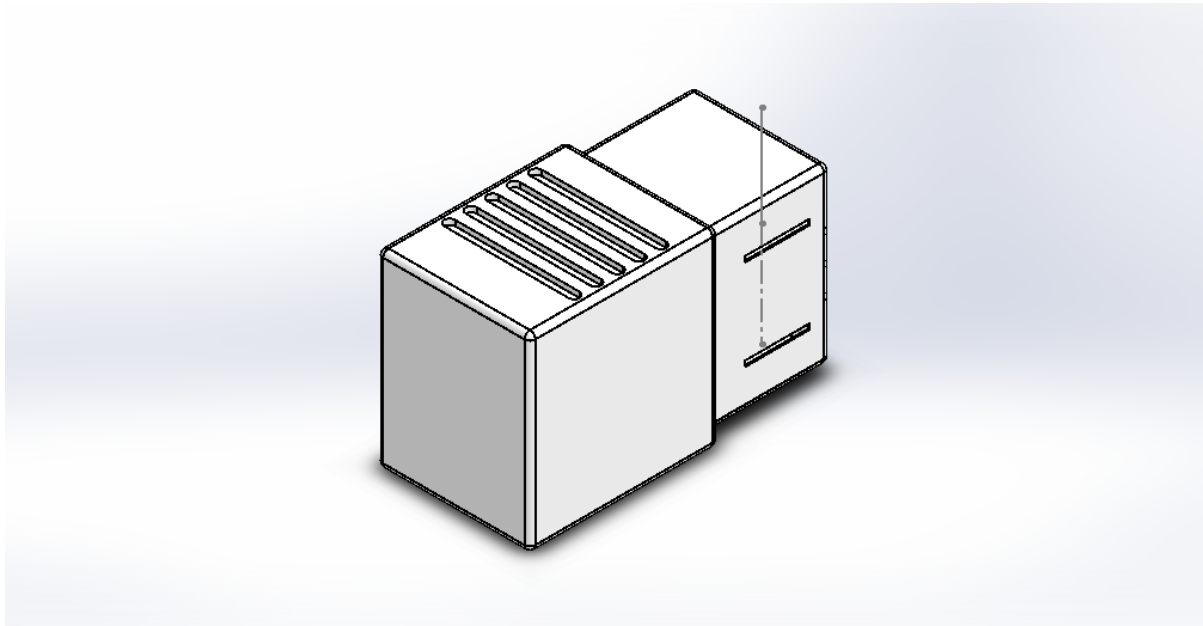
Technical sketch demonstrating fully defined geometry and complex relationships.

Symmetrical Base Layout



Geometric design for a high-precision 335mm span mounting base.

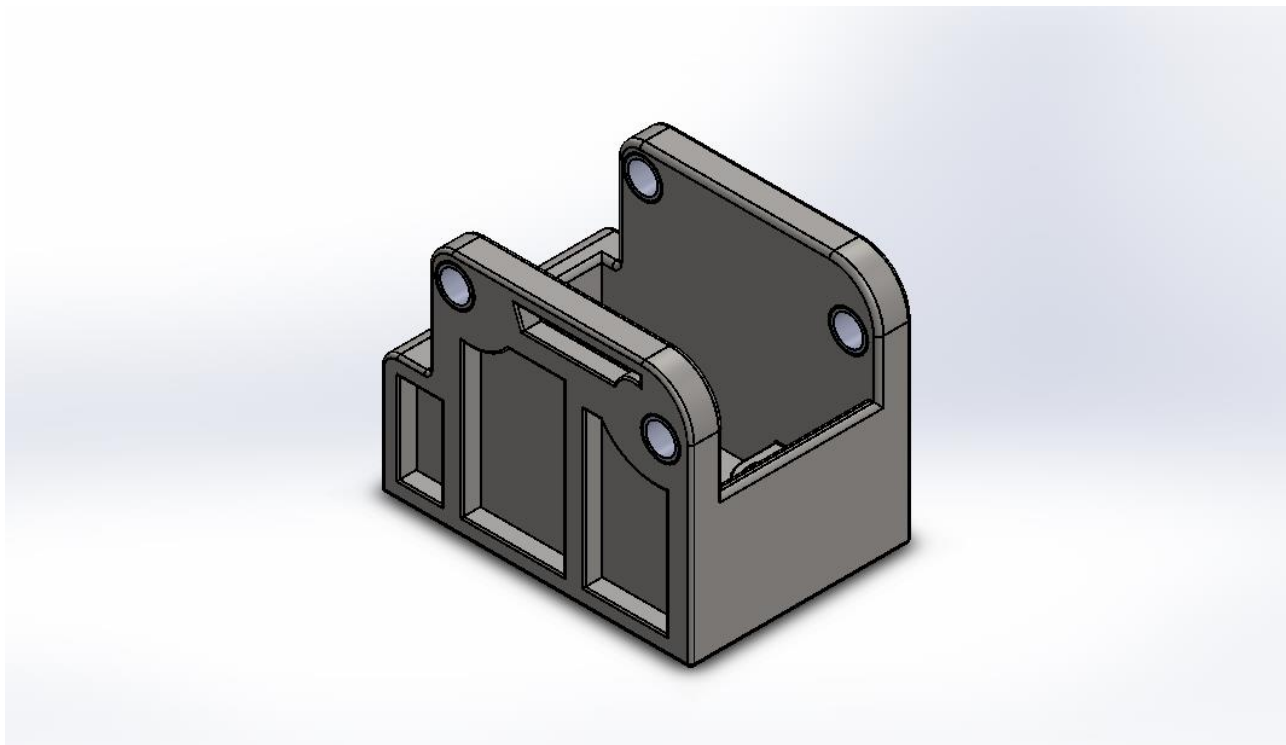
CNC Precision Machined Part



High-precision component designed specifically for CNC manufacturing processes, showcasing clean geometry and localized features.

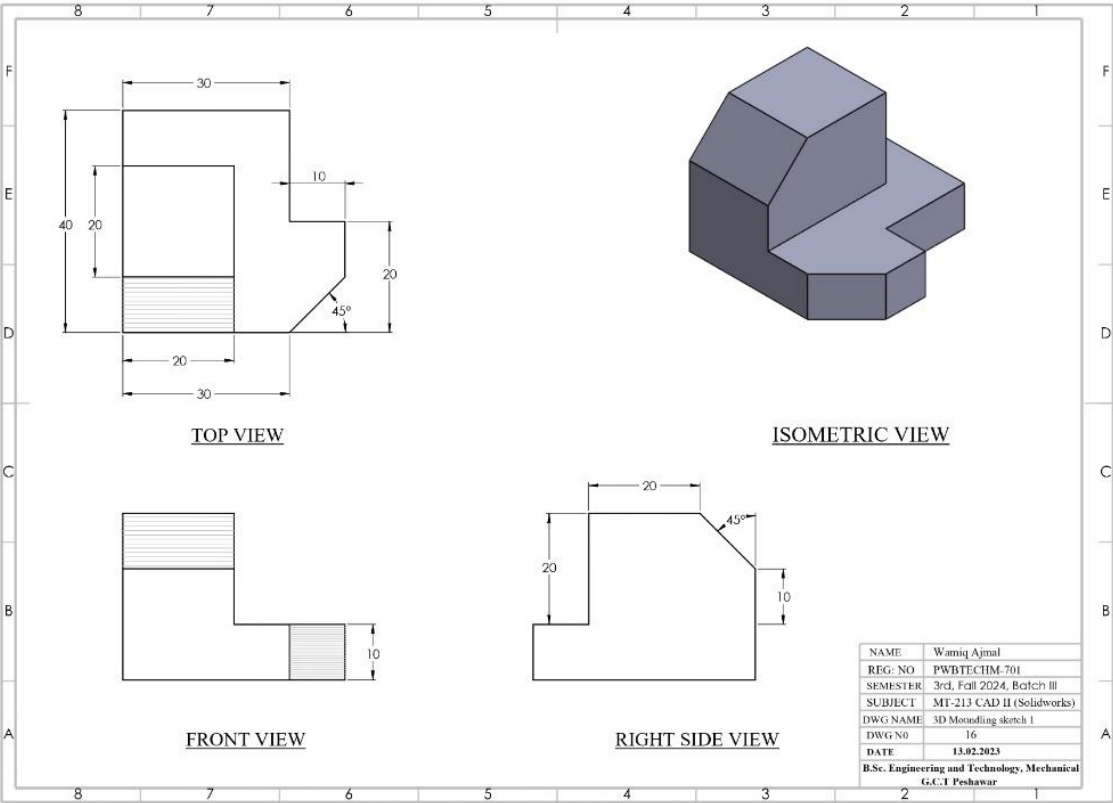
Industrial Assembly & Documentation

Mechanical System Assembly



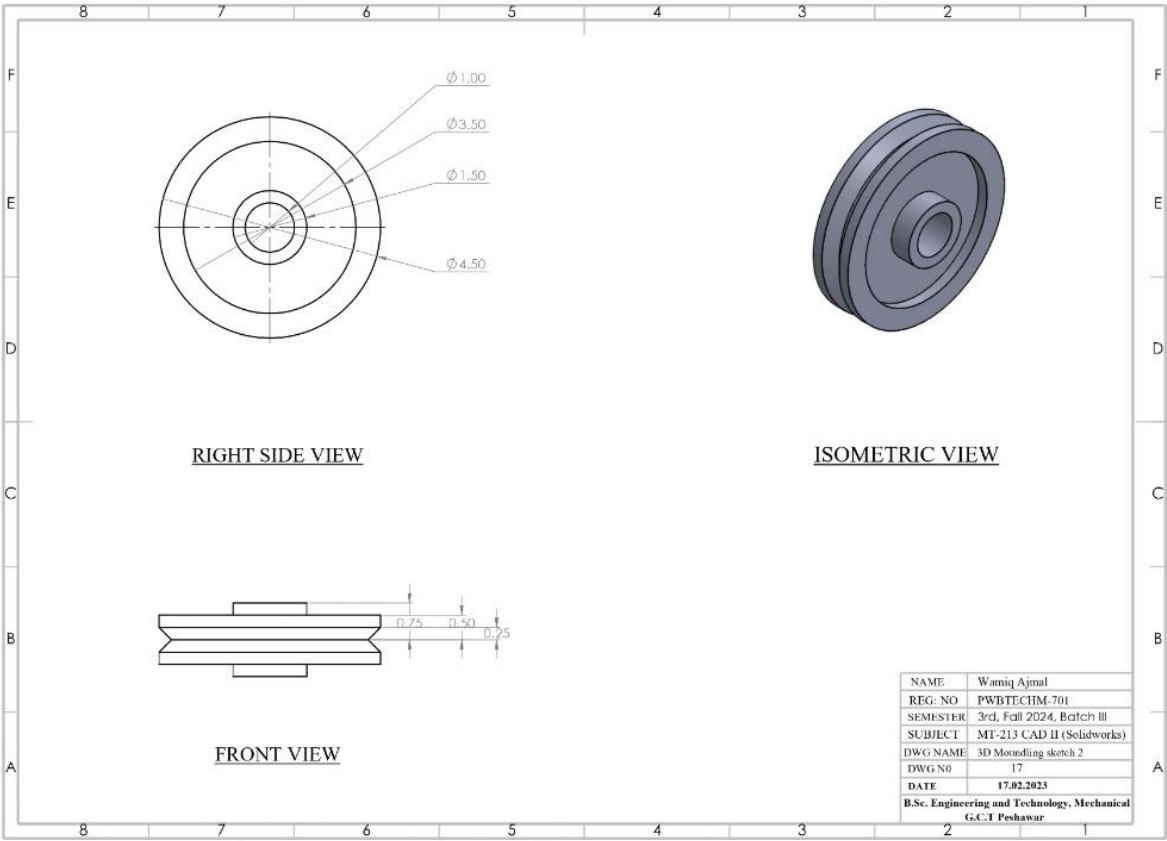
Full assembly view demonstrating part integration, interference checking, and mechanical relationship management.

Technical Engineering Documents



Formal engineering documentation including standardized title blocks, material specifications, and revision history.

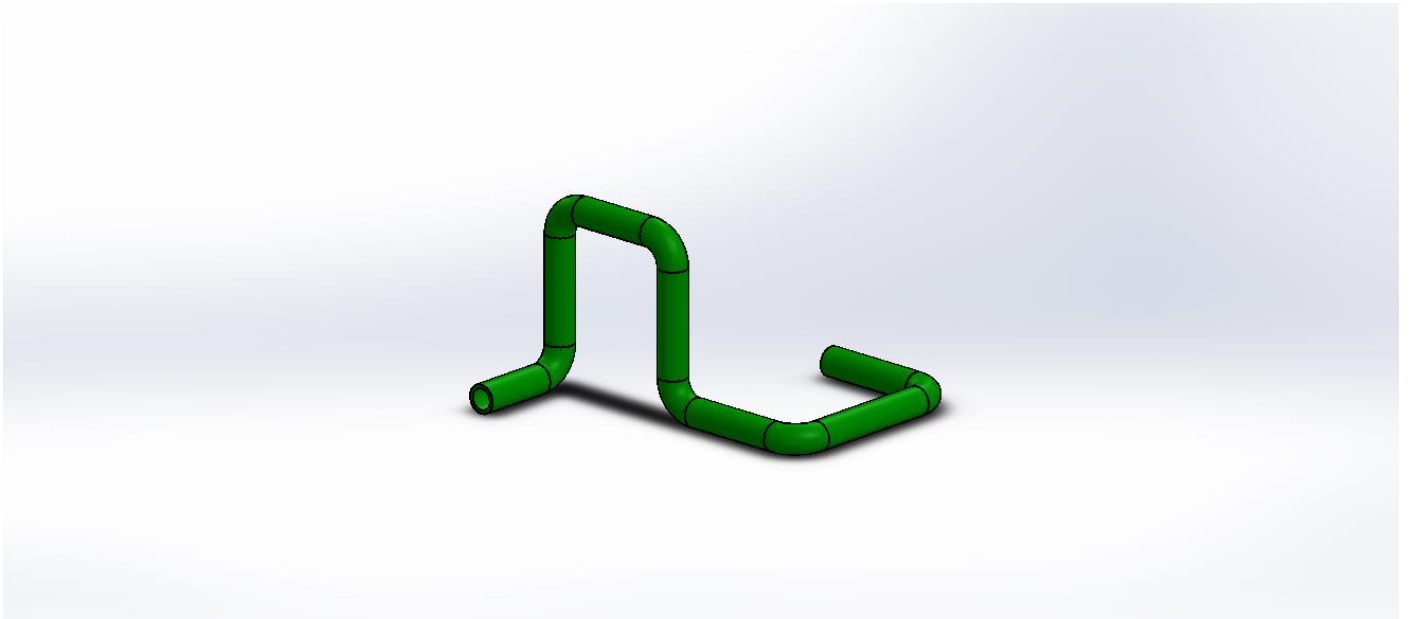
Industrial Fabrication Guide



Detailed manufacturing instruction sheet for assembly line integration and quality control.

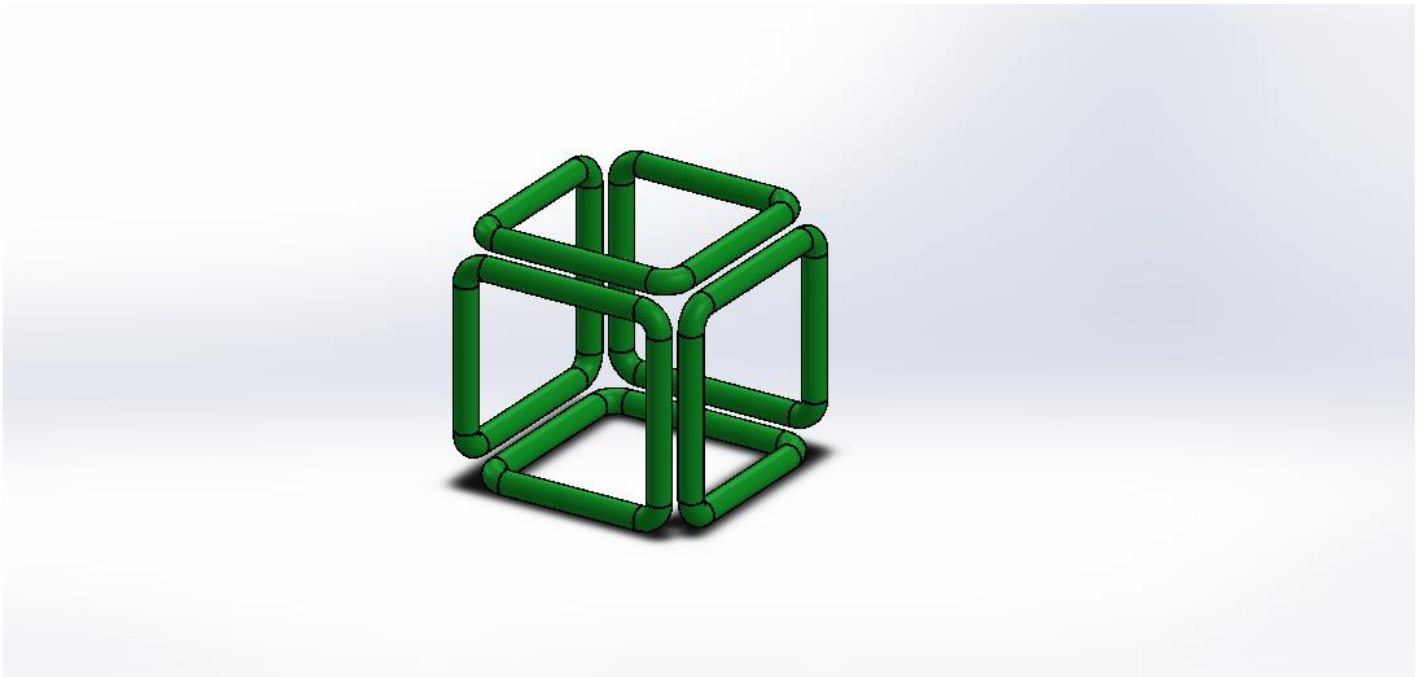
3D Routing & Wireframe Design

Complex 3D Routing Pattern



Demonstration of 3D sketch paths for complex hydraulic or pneumatic piping layouts.

Wireframe Structural Layout



3D parametric framework for structural spatial analysis and weldment prep.